

NUMERIC CONTINUITY SUBSTRATE FOR STRUCTURALLY NON-SEMANTIC PERSISTENCE OF LONGITUDINAL CONTINUITY

FIELD OF THE INVENTION

The present invention relates generally to computational systems and methods for maintaining longitudinal continuity across digitally mediated processes. More specifically, the invention relates to a non-semantic, numeric continuity substrate that enables persistence, re-entry, and continuity operations across time while being structurally incapable of storing, encoding, resolving, or deriving semantic meaning, personal data, identity, authorship, or interpretive content.

BACKGROUND OF THE INVENTION

1. Structural Conflation of Continuity and Semantics in Prior Art

Existing digital systems implicitly assume that longitudinal continuity requires the persistence of semantic representations of prior interaction or state. Such representations may include stored text, embeddings, inferred intent, preferences, profiles, reconstructed context, or other meaning-bearing artifacts retained across sessions.

Under this assumption, continuity is achieved only by persisting information that is interpretable in principle. This conflation of continuity with semantics is treated as foundational in prior systems.

The present invention rejects this assumption. Semantic persistence is not a necessary condition for continuity.

2. Inevitable Exploitation Under Semantic Persistence

When continuity is implemented through semantic representations, the retained state is inherently interpretable, inspectable, and economically extractable. This creates a persistent control surface for profiling, optimization, personalization, persuasion, or monetization that arises from the structure of the representation itself rather than from any particular operator intent.

Accordingly, systems that persist semantic continuity cannot, by design, render such continuity immune to repurposing or exploitation.

3. Insufficiency of Governance-Based Safeguards

Attempts to mitigate the risks of semantic continuity through encryption, access control, anonymization, minimization, or consent mechanisms do not

eliminate the underlying structural property: semantic meaning is retained and may be resolved in principle.

As a result, governance-based or policy-based safeguards are insufficient to make semantic continuity structurally safe.

4. Continuity as a Required System Function

Independent of how interaction is presented, modern computational systems increasingly require continuity across time. Such continuity includes, without limitation, the ability to resume prior activity, re-enter ongoing processes, coordinate across sessions, maintain temporal relationships, and preserve longitudinal structure.

As interaction is increasingly mediated through conversational, assistive, or agent-based interfaces, continuity becomes a prerequisite for human agency rather than a convenience.

5. Need for a Non-Semantic Continuity Primitive

There exists a need for a computational primitive that provides longitudinal continuity while being structurally incapable of storing, resolving, or exposing semantic meaning, identity, personal information, or interpretive content.

SUMMARY OF THE INVENTION

The present invention provides a numeric continuity substrate that persists longitudinal continuity across interaction sessions while being structurally incapable of encoding semantic meaning, personal data, identity, authorship, or interpretive content.

Continuity within the invention is represented exclusively as time-ordered numeric structures (“traces”) supplied by one or more external processes. The substrate does not interpret, infer, classify, or assign meaning to these structures and operates solely on numeric continuity coordinates and temporal relationships.

The continuity substrate may serve as a foundational component of an Intelligent Augmented Memory (IAM) system, in which computation does not infer or replace human judgment, but instead preserves the structural conditions under which human thinking can persist, be resumed, and evolve across time.

Traces may originate from any external source, including human-generated interactions, automated agents, tools, sensors, or system processes. Human-generated

traces represent a non-exclusive subclass and are not structurally privileged within the substrate.

The invention enables continuity operations including, without limitation, persistence across sessions, re-entry into prior continuity regions, recurrence detection, drift or divergence measurement, and stability or fragmentation analysis.

By rendering continuity state non-semantic and non-interpretable, the invention transforms continuity from an exploitable memory asset into a bounded computational resource whose structure may be reused without exposing or retaining meaning.

DETAILED DESCRIPTION OF THE INVENTION

1. Overview

The invention defines a foundational continuity layer that may be incorporated into higher-level systems while remaining strictly non-semantic. The layer persists continuity structure across time while enforcing an invariant that semantic meaning cannot be stored, inferred, or resolved within the substrate.

2. Numeric Continuity Substrate (NCS)

The Numeric Continuity Substrate (NCS) comprises a storage and computation layer configured to accept and persist discrete continuity elements (“continuons”).

Each continuon consists of:

- a numeric continuity coordinate set,
- a temporal position or ordering index, and
- an opaque numeric identifier.

All values persisted within the substrate are numeric and structurally incapable of encoding semantic meaning.

3. Trace Definition and Source Agnosticism

A trace is defined as a time-ordered numeric sequence supplied to the substrate by an external process. The substrate does not encode, infer, or retain the source, intent, authorship, or interpretation of any trace.

Human-generated traces, agent-generated traces, and system-generated traces are structurally indistinguishable within the substrate.

4. Structural Invariants

The substrate enforces the following invariants:

- no semantic content is stored;
- no persisted value is resolvable to meaning, identity, authorship, or personal data within the substrate; and
- no interpretation, classification, or inference is performed on stored values.

These invariants are enforced at the architectural level.

5. Separation from Semantic Systems

Semantic interpretation, language processing, or meaning assignment may occur, if at all, in systems external to the substrate. Any such processing is ephemeral and does not alter or persist within the continuity state stored by the substrate.

6. Continuity Access Protocol (CAP)

Access to the substrate is mediated by a Continuity Access Protocol (CAP) that exposes only structural operations over continuity state.

CAP permits:

- insertion of numeric continuity data,
- retrieval of continuity neighborhoods,
- traversal of continuity chains, and
- detection of recurrence or re-entry signals.

7. Continuity Operations Without Meaning

Using only numeric continuity data, the substrate supports computation of:

- persistence across sessions,
- recurrence of prior continuity regions,
- drift or divergence over time,
- stability or fragmentation of continuity segments, and
- re-entry into prior continuity neighborhoods.

Re-entry enables resumption of an ongoing process without reconstructing semantic context.

METHOD OF MAINTAINING LONGITUDINAL CONTINUITY

A method for maintaining longitudinal continuity across interaction sessions without semantic persistence, the method comprising:

- (a) receiving numeric continuity elements corresponding to activity occurring over time;
- (b) persisting the numeric continuity elements as time-ordered continuity state while structurally preventing storage of semantic meaning, identity, authorship, intent, or interpretive content;
- (c) enforcing an architectural invariant prohibiting semantic interpretation; and
- (d) enabling re-entry into a prior continuity region using numeric and temporal relationships.

ADVANTAGES OF THE INVENTION

The invention provides, among other advantages:

- persistence of continuity without semantic storage;
- structural immunity to profiling and manipulation; and
- suitability as a foundational continuity primitive.

CONCLUSION

The invention introduces a computational primitive for persistent longitudinal continuity that is structurally incapable of encoding meaning. By decoupling continuity from semantics at the architectural level, the invention preserves human agency while enabling durable continuity across time.